

	Case Name: Data Loss Prevention System	Sector	IT	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule Risk Analysis	Schedule with resources Schedule with costs Random simulation One of nine std. scenarios User defined distributions	
Submitted by	Mieke De Saedeleer	Project Control	Automatic tracking	
Date	2016		Tracking based on user input	
File Name	C2016-9 Data Loss Prevention System			

1. Project description

Project authenticity

The installation and implementation of a Data Loss Prevention (DLP) system within an organization. The project consists of activities such as kickoff meetings, workshops, documentation analysis, system design, and deployment, with associated resource and cost data derived from the actual project.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	143	Serial/Parallel (SP)	10%
Planned Duration (PD)	195	Activity Distribution (AD)	36%
Budget At Completion (BAC)	584,951.77 €	Length of Arcs (LA)	1%
Renewable Resources	-	Topological Float (TF)	51%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	10	9	1.42
CRI-rho	13	12	1.76
CRI-tau	11	21	3.77

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	NA	NA	NA
CRI-rho	NA	NA	NA
CRI-tau	NA	NA	NA

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	10	21	2.26
SI	28	26	1.13
SSI	3	7	3.98
CRI-r	11	9	1.98
CRI-rho	13	12	1.97
CRI-tau	13	21	3.53

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	46	-46
PV - SPI	25	15
PV - SCI	25	15
ED - 1	121	-121
ED - SPI	25	15
ED - SCI	25	15
ES - 1	68	-68
ES - SPI(t)	30	29
ES - SCI(t)	30	29

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.00	0.00
CPI	1	-1
SPI	13	13
SPI(t)	14	14
SCI	13	13
SCI(t)	14	14
0.8 CPI + 0.2 SPI	9	9
0.8 CPI + 0.2 SPI(t)	10	10

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 9 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-609,920.57	17366.34	8.52	0.45	1.02	1.04	0.82
std dev	322,765.36	28458.27	10.62	0.19	0.21	0.24	0.20
final	1.24	-0.43	0.00	2.71	-1.81	-2.30	-1.40

2.3.3.2. Time forecasting

PD	195
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Real Duration	189
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Late/early	3%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	189.67	9.47	3.53	-26.20
PV - SPI	204.00	67.34	15.34	-41.72
PV - SCI	463.89	57.66	145.44	-212.53
ED - 1	190.44	7.33	2.88	-26.85
ED - SPI	204.00	67.34	15.34	-41.72
ED - SCI	301.00	71.07	59.26	-113.63
ES - 1	186.67	10.75	4.17	-24.41
ES - SPI(t)	206.78	93.22	22.22	-45.73
ES - SCI(t)	297.22	95.45	59.73	-113.63

2.3.3.3. Cost forecasting

BAC	584,951.77 €
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Real Cost	1,425,155.96 €
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Over/under budget	144%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	1194872.65	322765.37	18.40	18.18
CPI	1421733.33	346075.44	17.31	0.27
SPI	1237975.44	247128.13	14.97	14.78
SPI(t)	1256145.56	253111.51	13.50	13.34
SCI	1443989.79	194047.76	11.20	-1.49
SCI(t)	1458268.01	184187.43	9.92	-2.61
0.8 CPI + 0.2 SPI	1326257.91	290689.27	14.28	7.80
0.8 CPI + 0.2 SPI(t)	1324595.39	290353.54	14.36	7.93